Unit 7

Campbell Biology, AP Edition, 12e

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Animal Form and Function

Steffanie Strathdee

Meet the "phage wrangler," Steffanie Strathdee, Professor of Medicine in the Division of Infectious Diseases and Global Public Health at the University of California, San Diego. Born in Canada, she completed her M.Sc. and Ph.D. degrees in epidemiology at the University of Toronto. Since joining the faculty at UC San Diego in 2004, she has focused on HIV research and prevention in the underserved populations of Tijuana, just across the Mexican border. In 2017, extreme personal circumstances prompted a radical shift in her research, as detailed in *The Perfect Predator: A Scientist's Race to Save Her Husband from a Deadly Superbug*, a book co-authored by Dr. Strathdee and her husband. In 2018, Dr. Strathdee became co-director of the newly established Center for Innovative Phage Applications and Therapeutics at UC San Diego.

An Interview with: Steffanie Strathdee

Tell us about your start in science.

I always had a natural curiosity as a young girl. I did not have an innate ability at science or math, even though I was interested in them. I struggled in math. Even in university, I got D's in calculus. I still don't like math, but I've learned to surround myself with people that do really well in math. I think it's important for

students to realize that just because they're not good at everything doesn't mean that they can't pursue their dreams.

How did you become interested in epidemiology?

After both my course instructor and my Masters and Ph.D. advisor passed away from AIDS, I decided I wanted to focus on ending the HIV epidemic. I had started out in the laboratory and realized I was really lousy at tissue culture experiments. So I turned my attention to public health, of which epidemiology is one part. Epidemiology involves studying risk factors and patterns, not just at an individual level, like what people eat and how they behave, but also the social, political, and economic forces that drive those behaviors.

What changed your research focus?

My husband became ill with a superbug infection—an infection caused by a bacterium that is resistant to many antibiotics. As an infectious disease epidemiologist, you would think that I would have understood the global crisis we are facing in multidrug-resistant organisms. But it wasn't until it hit me on a personal level that I fully understood the threat.

How did the story of your husband's illness unfold?

My husband and I were in Egypt, and he became very ill. The doctor gave him IV antibiotics and said, "He'll be fine," but he wasn't. It turned out that a gallstone had blocked his bile duct, causing an abscess (cavity) to form, and a superbug moved into that abscess and multiplied. Tom's particular bacterium had 51 different antibiotic-resistance genes. It was winning its battle against the immune system, none of the standard antibiotic treatments were working, and my husband was dying. I did a literature search and found a hundred-year-old, forgotten cure based on bacteriophages. These viruses, called phages for short, attack bacteria, but not human cells. Each phage is specific for a particular bacterium. Phages inject their DNA into bacteria, turning them into phage factories and killing the bacteria in the process. I emailed Tom's doctor, who is a colleague of Tom's and mine, and he thought the idea was worth a shot. So, I had to go out and find people who had phages that were active against Tom's bacterium, and that felt harder than looking for a needle in a haystack. Luckily, labs at the Naval Medical Research Center in Maryland and at Texas A&M University dedicated themselves to the search and found phages that matched. The phages were grown, purified, and injected into Tom's body, a billion phages per dose. Three days later, Tom woke up from his coma.

Dr. Steffanie Strathdee's husband, Dr. Tom Patterson, holding an electron micrograph of his superbug (left), and Dr. Strathdee, with an artist's rendition of the bacteriophage that defeated it (right).

Tell us a bit more about your collaborators.

A Ph.D. student at Texas A&M worked around the clock and found the phages that made up the first infusion. This student was at a really low point in her studies, thinking, "I don't know what I'm going to do with my career, I don't know if what I do really matters." And then she ended up helping save the life of a total stranger, and it's jump-started a whole new field—phage therapy.

> ***"This student … ended up helping save the life of a total stranger, and it's jump-started a whole new field."***

What advice do you have for students starting to study biology?

Students should look for mentors with whom they share the same values. For me, that was what really turned my D in calculus and some of the troubles that I had in science into success. I found mentors who were encouraging and helped me identify my strengths. Knowing what your weaknesses and your strengths are is, I think, one of the big components to making a successful career.